$\Lambda \to p\pi^-$ in the ePIC B0 tracker — update: hit seeding, the 5 keV threshold, and Λs up to 275 GeV

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Abstract

This note updates the report of 3 November 2026 (`docs/report/report.pdf`, "the first report"), which used truth-seeded B0 tracks from EICrecon main `8b44cae` and Λs of 50–100 GeV. Three things changed since. (1) The realistic B0 seeding of EICrecon#3116 (triplets of B0 hits, no truth) was built and compared with truth seeding on the same events: it finds a seed for every accepted daughter, reconstructs every accepted proton, and gives 1–2 percentage points more fully reconstructed Λs (13.1 % instead of 12.0 % of the generated Λs in the new geometry) with the same mass peak. Its price is more tracks from secondaries, which matters only in the thick old geometry. (2) The 5 keV B0 threshold recommended in the first report is now the EICrecon default (EICrecon#3084). (3) New samples at 100–275 GeV show that although the probability that a Λ decays before the B0 falls from 48 % to 17 % ($p\pi^-$, per generated Λ), the early decays are reconstructed much better because the π^- is harder, so the fraction of fully reconstructed Λs stays roughly flat up to 275 GeV. One result is not understood: the truth-seeded efficiency in the old geometry is much higher with the #3116 build than with main `8b44cae`.

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1 What changed since the first report

Table 1: Changes w.r.t. the first report.

	first report (3 Nov)	this update (12 Nov)
EICrecon	main `8b44cae` (local build)	EICrecon#3116 branch at `1ca147b` = main of 2026-11-11 + #3116 (contains #2901 and #3084)
B0 seeding	truth only (#2901)	truth *and* hit seeding, both written by the same job
B0 threshold	default 10 keV; 5 keV by option	5 keV is the default (#3084, merged 2026-11-03)
eic-shell image	nightly of 2026-11-02	nightly of 2026-11-07 (its `epic-main` already has the new B0)
old geometry	epic `43f427e` = geometry of the container	epic `43f427e`, built locally (`epic_43f427e`); same commit
new geometry	epic `ec774d8` (local build)	unchanged (= epic tag 26.11.0)
Λ momentum	50–100 GeV	50–100 GeV (same simulated events, re-reconstructed) *and* 100–275 GeV (new)

The definitions of the first report are unchanged (accepted daughter: its own simulated B0 hits cover ≥ 3 layers; tracking efficiency = reconstructed / accepted daughters; acceptance $\times$ efficiency per $p\pi^-$ decay). Two are added: the hit-seeded "reconstructed daughter" (Sec. 2) and, as a cleaner denominator, "per generated Λ" (Sec. 4). All macros take a `seeding` argument (`"truth"`, default, or `"hit"`); `scripts/make_results.sh 07 08` produces every number of this note.

2 Hit seeding (EICrecon#3116)

2.1 Status and method

Realistic B0 seeding is not yet in EICrecon main. The first proposal, EICrecon#3028, was closed on 2026-11-05 in favour of #3116 (open); EICrecon#3029, which shows as merged, is a bot include fix merged into the branch of #3028, not into main. #3116 seeds B0 tracks with the ACTS triplet seeder: one hit in each of three different B0 stations, the hits rotated so that the B0 dipole field at the middle hit lies along z (the ACTS seeder assumes a solenoid-like field), minimum momentum across the field 1 GeV, and tracks with ≥ 3 measurements. There is no assumption on the vertex or the charge, so it works for displaced decays by construction.

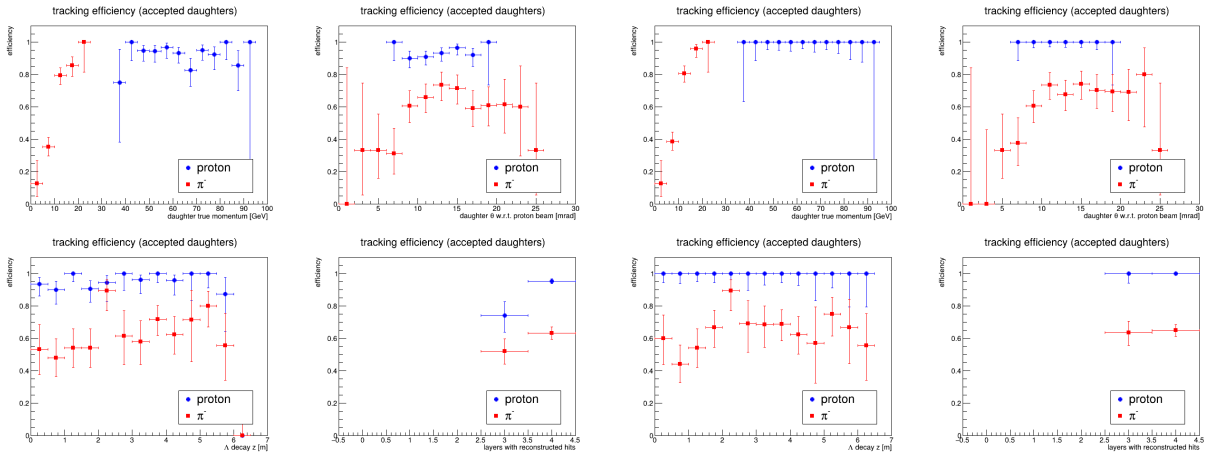
The chain is `B0TrackerSeeds` → `B0TrackerCKFTracks` (and `B0TrackerCKFTrack[Unfiltered]Associations`) → `CombinedTracks` → `ReconstructedChargedParticles` (+`Associations`). The truth-seeded chain of #2901 runs in parallel in the same job (`ReconstructedTruthSeededChargedParticles`), so the two are compared on identical events and hits. Hit seeds carry no MC link; a seed is assigned to a daughter if at least two of its three hits were made by it (via `B0TrackerRawHitAssociations`).

The 1000-event samples of the first report (50–100 GeV, seeds 101 and 102) were re-reconstructed with the #3116 build (`scripts/production/lambda_rereco.sh`); the simulation is the same.

2.2 Tracking efficiency

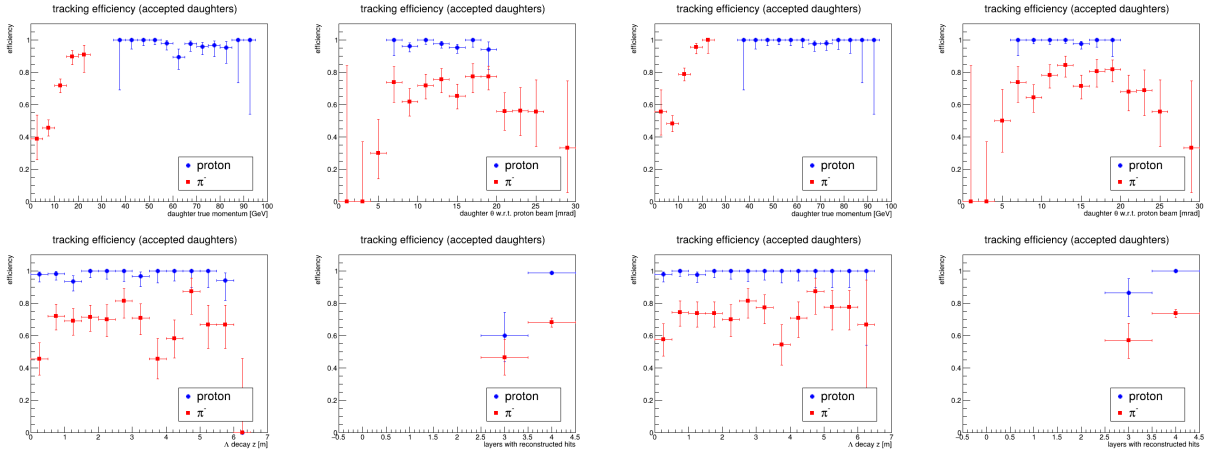
Table 2: Accepted daughters passing each step, proton / π^-, #3116 build, 5 keV, 1000 Λ per geometry.

step	old, truth	old, hit	new, truth	new, hit
accepted	404 / 349	404 / 349	295 / 240	295 / 240
rec. hits in ≥ 3 layers	100 / 100 %	100 / 100 %	100 / 100 %	100 / 100 %
seed	99 / 99 %	**100 / 100 %**	97 / 96 %	**100 / 100 %**
CKF track	98 / 66 %	100 / 74 %	94 / 61 %	100 / 67 %
filtered track	98 / 66 %	100 / 73 %	94 / 61 %	100 / 65 %
reconstructed particle	**97.5 / 66.5 %**	**99.5 / 72.5 %**	**93.2 / 60.8 %**	**100 / 64.6 %**



(a) new geometry, truth seeding

(b) new geometry, hit seeding



(c) old geometry, truth seeding

(d) old geometry, hit seeding

Figure 1: Tracking efficiency of accepted daughters (proton blue, π^- red) vs the daughter's true momentum, its θ, the Λ decay z and the number of layers with reconstructed hits.

- Hit seeding finds a seed for *every* accepted daughter; with truth seeding 1–4 % have no matching seed. With hit seeding every accepted proton is reconstructed.
- The π^- is still lost in the CKF (seed $\rightarrow$ track), just as with truth seeding: of the accepted π^- of 5–10 GeV only 39 % (new) and 48 % (old) are reconstructed with hit seeding, 35 % and 46 % with truth seeding. Since every one of them now has a seed made of its own hits, **the soft-π^- loss is not a seeding problem**.

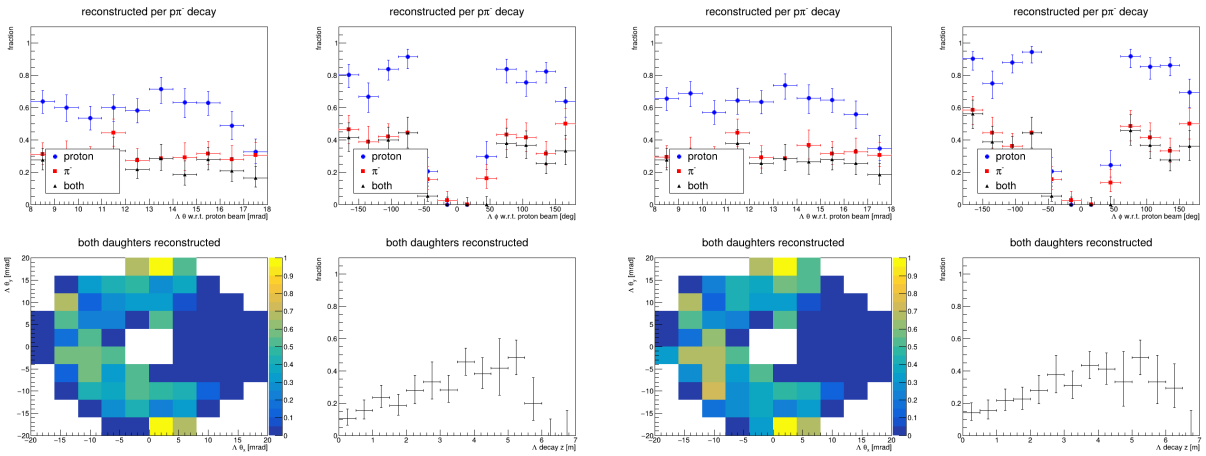
2.3 Acceptance × efficiency

Table 3: Acceptance × efficiency, first report vs this update. The acceptance is unchanged (same simulation): 80 / 69 / 63 % (old), 61 / 49 / 41 % (new), proton / π^- / both.

	first report: main `8b44cae`, truth	#3116 build, truth	#3116 build, **hit**
old geometry			
tracking efficiency, p / π^-	82.2 / 55.6 %	**97.5 / 66.5 %**	99.5 / 72.5 %
$A \times \varepsilon$ both, per $p\pi^-$ decay	29.8 %	**41.7 %**	45.5 %
both reconstructed, per generated Λ	15.0 %	21.0 %	**22.9 %**
new geometry (first report: 5 keV by option)			
tracking efficiency, p / π^-	92.9 / 60.8 %	93.2 / 60.8 %	100 / 64.6 %
$A \times \varepsilon$ both, per $p\pi^-$ decay	24.7 %	24.7 %	27.0 %
both reconstructed, per generated Λ	12.0 %	12.0 %	**13.1 %**

New geometry. Truth seeding with the #3116 build reproduces the first report exactly (the 5 keV default of #3084 equals the 5 keV option used then). Hit seeding adds 11 fully reconstructed Λs per 1000 (12.0 → 13.1 %).

Old geometry: not understood. With the same simulated events, truth-seeded tracking in the old geometry is much more efficient with the #3116 build than with main `8b44cae` (proton 82.2 → 97.5 %, both daughters per generated Λ 15.0 → 21.0 %), while the new geometry does not move. Three things differ between the two reconstructions: EICrecon (`8b44cae` → `1ca147b`: 10 days of main, plus #3116), the eic-shell image (ACTS and other libraries, nightly of 2026-11-02 → 11-07), and the epic build of the old geometry (the container's copy → a local build of the same commit). The threshold is not the cause (it makes no difference in the 300 μm geometry). This has not been investigated yet; until it is, old-geometry numbers of the two reconstructions should not be compared with each other.



(a) new geometry, truth seeding

(b) new geometry, hit seeding

Figure 2: Acceptance × efficiency per $p\pi^-$ decay vs the Λ θ, ϕ, (θ_x, θ_y) and decay z (proton blue, π^- red, both black).

3 The Λ mass with hit-seeded tracks

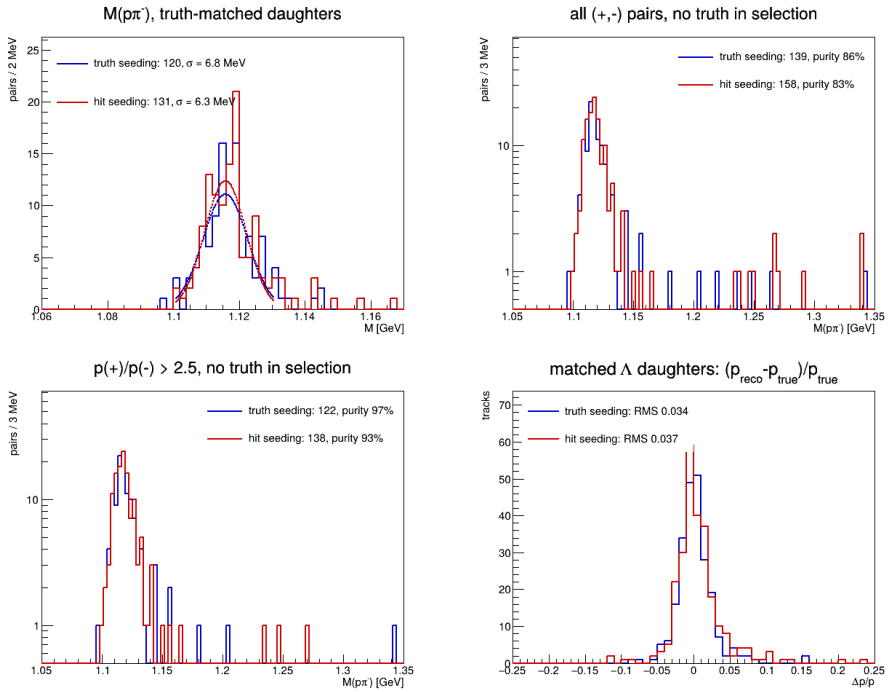


Figure 3: New geometry: truth seeding (blue) vs hit seeding (red), same events (`lambda_mass_seeding_compare.C`). Top: truth-matched $M(p\pi^-)$ with Gaussian fits; all $(+,-)$ pairs (log scale). Bottom: pairs after $p(+)/p(-) > 2.5$; $\Delta p/p$ of the matched daughters.

Table 4: Mass and selection without truth, truth vs hit seeding (#3116 build, 1000 Λ per geometry).

	old, truth	old, hit	new, truth	new, hit
truth-matched pairs	210	229	120	131
Gaussian mean [GeV]	1.1174	1.1175	1.1159	1.1159
Gaussian σ [MeV]	6.9	6.8	6.8	6.3
$\Delta p/p$ of the daughters, RMS	3.8 %	4.4 %	3.4 %	3.7 %
reconstructed tracks ("other" = not a Λ daughter)	824 (198)	1097 (432)	496 (68)	548 (91)
all $(+,-)$ pairs / purity	331 / 63 %	877 / 26 %	139 / 86 %	158 / 83 %
after $p(+)/p(-) > 2.5$ / purity	242 / 86 %	348 / **64 %**	122 / 97 %	138 / **93 %**
±20 MeV window: signal / candidates	200 / 207	206 / 219	110 / 110	116 / 116

- **Same peak.** For the decays reconstructed with both seedings (old 209, new 114), $M(\text{hit}) - M(\text{truth})$ has mean ≈ 0 and RMS 4.4 MeV. Hit seeding adds 17–20 decays and loses 1–6. The momentum resolution is slightly worse with hit seeding (3.7 vs 3.4 % in the new geometry): the fitted tracks differ a little.
- **More background.** Hit seeding also builds tracks for secondaries from interactions in material; truth seeding only seeds charged MC particles produced inside its window (vertex $z < 6$ m, $|x|, |y| < 240$ mm), presumably the reason for the difference (not checked). In the old geometry, with 300 μm sensors, this doubles the "other" tracks and the purity after the momentum-ratio cut drops from 86 to 64 %; in the new geometry the effect is small (97 → 93 %). The ±20 MeV window stays ≥ 94 % pure in all cases.
- The +2 MeV shift of the old geometry is still there with both seedings.

4 Sample composition: the right denominator

The first report stated that 88–90 % of the $p\pi^-$ decays happen before the B0 tracker ($z < 5.8\,\mathrm{m}$), and that the 24–25 % of Λs with fate "other" interact in material *or leave the world volume*. The second part is wrong, and the first needs a comment. The fates of the 1000 Λs of the new-geometry sample (50–100 GeV):

fate	number	end at $z < 5.8\,\mathrm{m}$	last end z
decay to $p\pi^-$	485	438	8.7 m
decay to $n\pi^0$	278	246	9.4 m
other	237	11	12.4 m

All 237 "other" Λs have daughters, i.e. they *interacted*; 205 of them end at $z = 6$–10 m. A Λ that has not decayed by about 6 m nearly always interacts in the material downstream before it can decay. The $p\pi^-$ decays that exist are therefore mostly early ones, and "90 % before the B0" is a fraction of the $p\pi^-$ decays, not of the Λs. Per generated Λ the exponential decay law holds: 684 of 1000 (68 %) decay before 5.8 m, against 66 % from $1 - e^{-L/\lambda}$ averaged over 50–100 GeV, and 43.8 % decay to $p\pi^-$ there, against $0.641 \times 66\,\% = 42\,\%$.

Consequence: results "per $p\pi^-$ decay" (acceptance, $A \times \varepsilon$) are conditional on the decay having happened in the simulated region. "Per generated Λ" is the unbiased figure and is used below.

5 Λs up to 275 GeV

5.1 Samples and expectation

Two new samples per geometry, 500 events each (seeds 201, 202), with the same gun but $p =$ 100–275 GeV (`PMIN=100 PMAX=275 lambda_sample.sh`, simulation only, then `lambda_rereco.sh` with the #3116 build). Together with the 50–100 GeV samples they cover 50–275 GeV; `lambda_vs_momentum.C` reads both.

The mean decay length is $\lambda = (p/m)\,c\tau = p[\mathrm{GeV}] \times 7.07\,\mathrm{cm}$ ($c\tau = 7.89\,\mathrm{cm}$, $m = 1.1157\,\mathrm{GeV}$). The probability to decay before the B0 ($L = 5.8\,\mathrm{m}$; the path equals z to 10^{-4} at $\sim 10\,\mathrm{mrad}$) is $1 - e^{-L/\lambda}$: 81 % at 50 GeV, 56 % at 100 GeV, 34 % at 200 GeV and 26 % at 275 GeV. At beam-like momenta most Λs thus reach the B0 undecayed. On the other hand the π^- carries 6–25 % of p_Λ, so at high momentum it is no longer soft.

5.2 Results

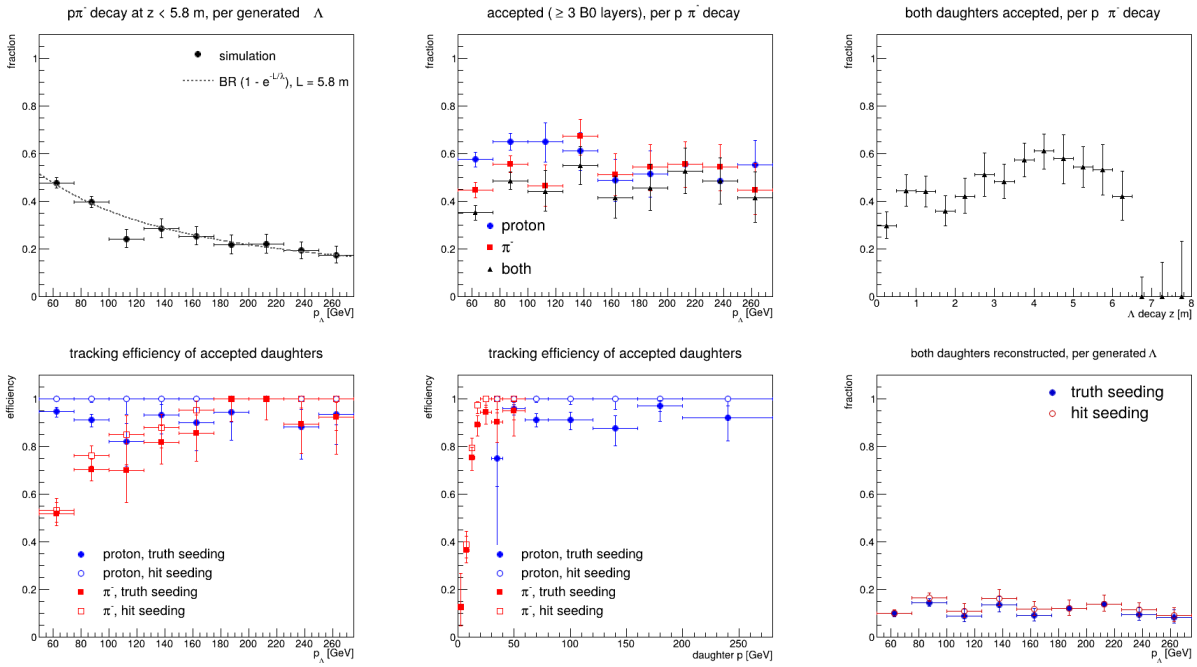


Figure 4: New geometry, 50–275 GeV, #3116 build. Top: $p\pi^-$ decays at $z < 5.8$ m per generated Λ with the decay-law expectation (dashed); daughter acceptance per $p\pi^-$ decay vs p_Λ; acceptance of both daughters vs decay z. Bottom: tracking efficiency of accepted daughters vs p_Λ and vs the daughter's own momentum (filled: truth seeding, open: hit seeding); both daughters reconstructed per generated Λ.

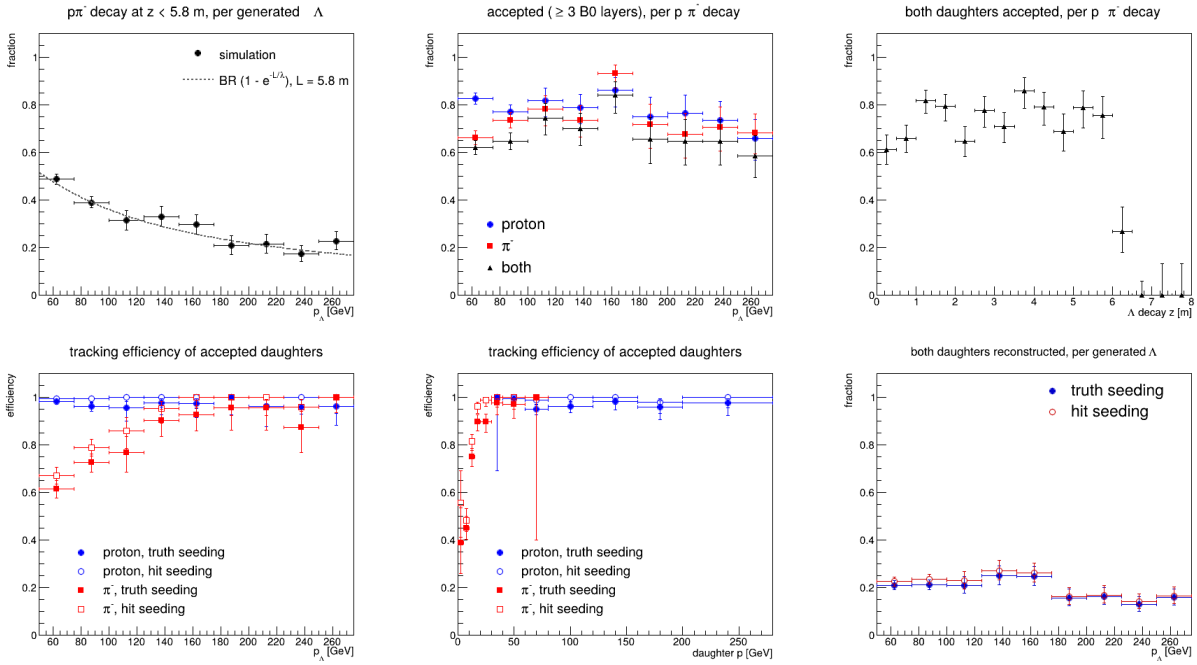


Figure 5: As Fig. 4, old geometry.

Table 5: Per 25 GeV bin of p_Λ (about 500 Λ per bin below 100 GeV, 130–155 above): $p\pi^-$ decays at $z < 5.8$ m per generated Λ (simulated / decay law), both daughters accepted per $p\pi^-$ decay, tracking efficiency of accepted p and π^- (truth / hit seeding), and both daughters reconstructed per generated Λ (truth / hit).

	old geometry						new geometry					
p_Λ [GeV]	decay	law	acc. both	ε_p	ε_π	**both rec.**	decay	law	acc. both	ε_p	ε_π	**both rec.**
50–75	49	47	62	98/100	62/67	21/23	48	47	35	95/100	52/53	10/10
75–100	39	39	65	96/99	73/79	21/23	40	39	49	91/100	70/76	14/17
100–125	31	33	75	96/100	77/86	21/23	24	33	44	82/100	70/85	9/11
125–150	33	29	70	98/100	91/95	25/27	28	29	55	93/100	82/88	14/16
150–175	30	25	84	97/100	93/100	25/26	25	25	42	90/100	86/95	9/12
175–200	21	23	66	100/100	96/100	15/16	22	23	46	94/100	100/100	12/12
200–225	21	21	65	96/96	96/100	16/17	22	21	53	100/100	100/100	14/14
225–250	17	19	65	96/100	88/96	13/14	19	19	49	88/100	90/100	9/11
250–275	23	17	60	96/100	100/100	16/17	18	17	43	94/100	93/100	9/10

All numbers in %. Above 100 GeV each bin has only 30–55 $p\pi^-$ decays, so the statistical errors are 5–10 percentage points.

- **Decay before the B0** follows the decay law over the whole range, 48 % → 17–18 % of the generated Λs (as $p\pi^-$).
- **Tracking efficiency rises.** The π^- efficiency vs its own momentum goes from ≈ 40 % below 5 GeV to ≈ 100 % above ≈ 20 GeV for both seedings; vs p_Λ it rises from 52–67 % at 50–75 GeV to 90–100 % above 150 GeV. Protons: 100 % with hit seeding at all momenta; truth seeding loses 5–10 % in the new geometry.
- **Acceptance** of both daughters per $p\pi^-$ decay stays at 60–85 % (old) and 35–55 % (new); in the new geometry it is limited by the sensor coverage (ϕ gap), not by the momentum. Decays beyond $z \approx 6.5$ m are never accepted (fewer than three layers left).
- **Net effect: roughly flat.** Both daughters reconstructed per generated Λ: old 21–27 % up to 175 GeV and 13–17 % above; new 9–17 % over the whole range. The falling decay probability is compensated by the better reconstruction of the early decays: in the new geometry 21 % of the early $p\pi^-$ decays are fully reconstructed at 50–75 GeV, about 40–60 % above 150 GeV.

5.3 Differences between the 50–100 and 100–275 GeV samples

Same gun (apart from the momentum), same geometries at the same commits and the same reconstruction (#3116 build) for both. The simulation, however, ran in different eic-shell images (2026-11-02 and 11-07), and the old geometry came from the container in the first case and from a local build of the same commit in the second; given the unexplained old-geometry difference above, the old-geometry curves across 100 GeV carry that caveat. The new-geometry results were stable under the same changes. The 100–275 GeV files are about three times larger per event (1.2–1.5 GB per 500 events), presumably from the larger showers of energetic Λs interacting downstream (not checked).

6 Summary and open points

- **Hit seeding (EICrecon#3116) works for displaced Λ daughters:** a seed for every accepted daughter, every accepted proton reconstructed, 1–2 percentage points more fully reconstructed Λs than truth seeding (new geometry 12.0 → 13.1 % of the generated Λs), and the same mass peak ($\sigma \approx$ 6–7 MeV). It also tracks secondaries; without truth the purity after the momentum-ratio cut is 93 % (new) and 64 % (old geometry).
- **The soft-π^- loss is in the CKF, not in the seeding**: it persists with a seed for every π^-.

- **5 keV** is now the default B0 threshold (#3084).
- **Energetic Λs:** fewer decay before the B0 (decay law), but those that do are reconstructed better; the fully reconstructed fraction is roughly flat up to 275 GeV.
- **Denominator:** "per generated Λ" is the unbiased figure; "per $p\pi^-$ decay" is conditional on an early decay.

Open points: the jump of the old-geometry truth-seeded efficiency between EICrecon builds (rerun `8b44cae` in the new image, or `1ca147b` with the container's old epic); the CKF loss of soft tracks; the +2 MeV mass shift in the old geometry; following #3116 until it is merged; events with other particles and beam background, where the decay vertex will be needed.

A Reproducing

- Builds: `scripts/build/build_eicrecon.sh EICrecon_pr3116 3116` (a PR) and `scripts/build/build_epic.sh epic_43f427e <full commit>` (a commit). `GEO` and `RECO` of the scripts accept a directory name under `\$EIC_TOP` or `/opt/detector`.
- Re-reconstruction: `scripts/production/lambda_rereco.sh <GEO> <NAME> EICrecon_pr3116 eicreconpr3116 [EXTRA] [DEST]`.
- High-momentum samples: `PMIN=100 PMAX=275 SIM_ONLY=1 scripts/production/lambda_sample.sh ...`.
- Results: `scripts/make_results.sh 07 08`. Files on EOS: `lambda_1000ev/*.eicreconpr3116.*`, `lambda_100to275/`.
- Pitfall: JANA keeps the *first* value of a repeated `-P` option, so `-Pjana:nevents=20` after `-Pjana:nevents=-1` is ignored.